

Anirudh Maiya

None, None

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Summary

Innovative Software and AI Engineer with a strong foundation in Computer Science and a proven track record of taking complex technical solutions from research to production. Expert in designing scalable cloud-native architectures, optimizing large-scale data processing systems, and developing novel machine learning frameworks. Experienced in migrating legacy systems to modern Python environments, implementing robust DevOps practices, and driving significant improvements in system reliability and performance. Adept at bridging the gap between cutting-edge AI research and practical, high-impact software applications within enterprise ecosystems.

Work Experience

- Commvault** None, None
Software Engineer *January 2022 - August 2023*
 - Elevated remote troubleshooting service success rate from 85% to 98% by improving software robustness and implementing comprehensive error handling.
 - Led a major migration and refactoring of a 35,000+ line remote troubleshooting service, upgrading the codebase from Python 2.7 to Python 3.9.
 - Replaced legacy Airflow and Luigi dependencies with a high-performance in-house Python codebase, implementing OOP principles and generic retry APIs.
 - Revamped the Commvault Plugin for ServiceNow to ensure full compliance with ServiceNow Store regulations and acted as the SME for workflow integrations.
 - Managed critical library migrations for cloud services Java projects, including jQuery and Tomcat upgrades.
 - Streamlined code adaptability for production and test environments through integrated dependency management and log scrubbing.
- Commvault** None, None
Associate Software Engineer *July 2021 - December 2021*
 - Designed and developed 'Decompress-as-a-service' within the cloud ecosystem, implementing a leader-worker architecture.
 - Enabled over 1,000 engineers to process 30+ terabytes of customer logs daily through optimized aggregation and decompression functionalities.
 - Achieved annual time savings of 300 hours for the Cloud-Services Team by implementing on-the-fly maintenance and feature release upgrades.
 - Managed workflows for the Commvault Cloud platform serving 20,000+ customers and mitigated security vulnerabilities by decoupling licensing-dependent reports.
- Commvault** None, None
Development Intern *January 2021 - June 2021*
 - Optimized cloud services build performance by migrating dependency management from Apache Ant to Gradle, reducing build times from 3 minutes to 30 seconds.

- Enabled rapid response to the Log4j vulnerability through improved build orchestration.
- Designed and automated comprehensive test cases for the Commvault Cloud Disaster Recovery solution.

- **Indian Space Research Organization**

None, None

Project Intern

October 2019 - April 2020

- Developed a novel framework for estimating coconut farm area using multi-spectral data, achieving 89% accuracy.
- Introduced a framework facilitating contradictory feature learning for enhanced precision in remote sensing.
- Presented research findings at the IEEE IGARSS - 2021 conference.

Projects

- **Tom Optimizer**

Researcher

Now

- Developed a novel variant of the Adam optimizer that accounts for gradient trends.
- Achieved an average improvement of 4.6% in accuracy and convergence speed compared to traditional optimizers.

- **Cloud Image Prior**

Researcher

Now

- Devised a training-data-free framework to remove cloud cover from Sentinel-2 remote-sensing imagery.
- Boosted average peak signal-to-noise ratio (PSNR) by 64%.

- **Messup**

Researcher

Now

- Created a novel regularization technique using exponential smoothing for sample encounters.
- Reduced generalization error by 1.6% on the CIFAR-10 dataset when integrated with ResNet.

- **RideShare Application**

Backend Developer

Now

- Built a scalable backend for a ride-sharing platform using RabbitMQ for message queuing.
- Orchestrated worker containers and implemented eventual consistency for database updates.
- Ensured high system fault tolerance using Apache ZooKeeper.

Education

- **University of Colorado Boulder**

None, None

Computer Science

Now - May 2025

- **PES University**

None, None

Computer Science

Now - May 2021

Technical Skills

Python, Java, SQL, PostgreSQL, Microsoft SQL Server, MongoDB, PyTorch, TensorFlow, Keras, Scikit-learn, Numpy, Pandas, Matplotlib, Plotly, Hugging Face, NLTK, SpaCy, REST API, Flask, Node.js, ReactJS, JavaScript, HTML, CSS, JSON, XML, Docker, Kubernetes, Azure, Google Cloud Platform, Apache Spark, RabbitMQ, Apache ZooKeeper, Gradle, Apache Ant, Git, Linux, Operating Systems, Algorithms, Computer Vision, Feature Learning, Multi-spectral Data, Disaster Recovery, Test Automation, ServiceNow Integration

Publications

- **Research on Coconut Farm Area Estimation using Multi-spectral Data (IEEE IGARSS - 2021)**

2021